Théo MAHIEU, Edgar ROUSSEL, Aurélie BECHINGER

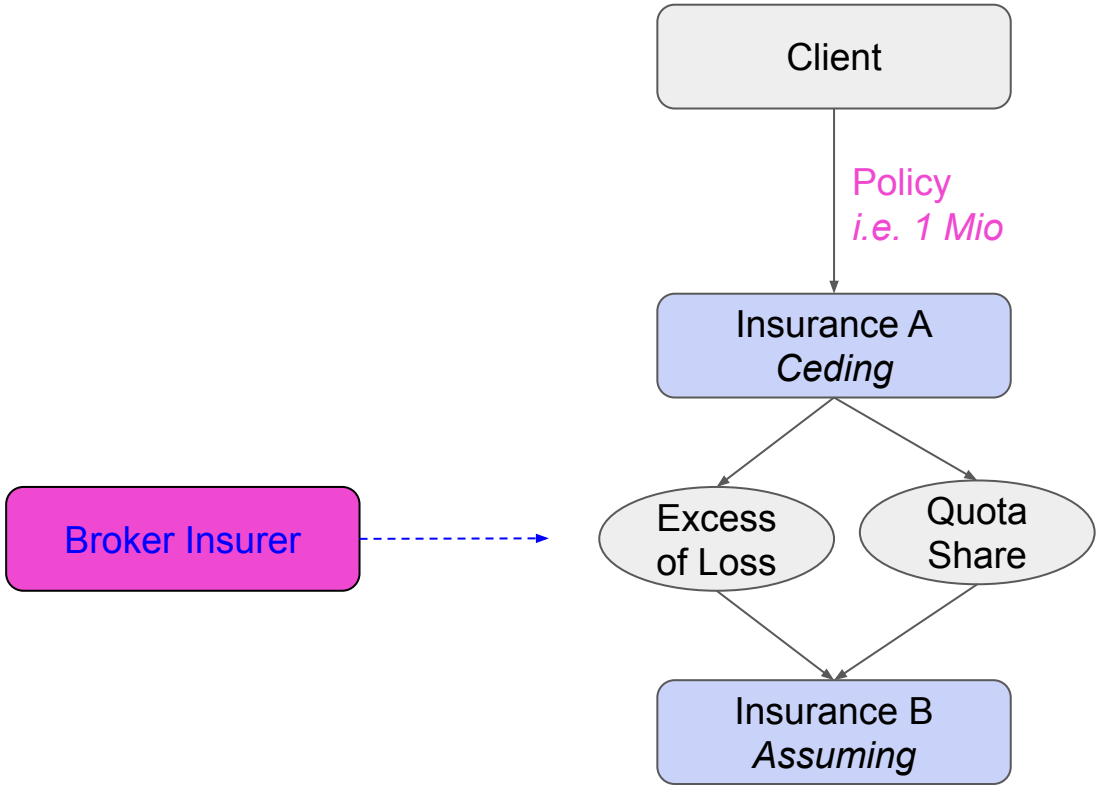
Quantum inspired reinsurance portfolio optimisation

quobly

AXA

QuobleStone

Reinsurance



EoL

Min: 0.5 Mio
Max: 2 Mio

Quota Ratio

30%

Incident: 10 Mio

From A: 8 Mio
From B: 2 Mio

From A: 7 Mio
From B: 3 Mio

1. Theoretical framework

Data set:

- Risk drivers: floods, fires, cyber attacks…: mathematical models
- Reinsurance contracts
- For each simulation s, calculate the recovery for each contract

Excess loss: R = min(max(L - deductible),0), limit)

Quota share: R = L x %share

Mathematical function: QUBO

$$C(x) = \underbrace{\sum_{i=1}^{n}(P_i - E[R_i])x_i}_{\text{Net Cost}} + \underbrace{\gamma \sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(R_i, R_j)_{tail} x_i x_j}_{\text{Tail Risk Variance}} + \underbrace{\alpha \left(\sum_{i=1}^{n} K_i x_i - S\right)^2}_{\text{Capital Constraint}}$$

Toward an Ising Hamiltonian:

$$H_C = \sum_{i=1}^{n} h_i Z_i + \sum_{i<j}^{n} J_{ij} Z_i Z_j$$

2. VQE approach

Warm start VQE:

- Hardware-efficient ansatz:
  - CNOTs for entanglement
  - Ry(θ) and Rz(ϕ) for rotating the qubits
- Initial state optimized by preliminary (meta-)VQE:
  - Encode Hamiltonian parameters (α in this case):

$$R(f(\alpha, \vec{\phi}_{ij})) = R_y(w_{ij}^{(1)}\alpha + \varphi_{ij}^{(1)})R_z(w_{ij}^{(2)}\alpha + \varphi_{ij}^{(2)}),$$

  - Obtain optimized angles for initial guess of VQE
- BFGS optimization

3. Why this approach?

- Normal VQE never converged to one specific value in reproducibility experiments
- Distribution of results was unintuitive
- Meta-VQE allowed for value convergence in reproducibility experiments
- Distribution is still a bit awkward

4. Advanced QAOA approach

Boosted QAOA:

- Different mixers: X, Cat Map, Fermionic
- Multi-angle $|\psi(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle = \prod_{l=1}^{p} \left(e^{-i\sum_k \gamma_{l,k} P_k} e^{-i\sum_i \beta_{l,i} X_i}\right) |+\rangle^{\otimes n}$
- Fourier parametrized angles $\gamma_l = \sum_{k=0}^{K} d_k \cos\left(\frac{2\pi kl}{p}\right) + f_k \sin\left(\frac{2\pi kl}{p}\right)$ $\beta_l = \sum_{k=0}^{K} q_k \cos\left(\frac{2\pi kl}{p}\right) + r_k \sin\left(\frac{2\pi kl}{p}\right)$
- Genetic optimisation

5. Risk reinsurance

- Risk models

```
# 1. Windstorm (Weibull): High-intensity peaks
windstorm = np.random.weibull(1.5, n_sims) * 40
# 2. Pluvial Flood (Pareto): The primary tail driver (Fat Tail)
flood = (np.random.pareto(1.2, n_sims) + 1) * 60
# 3. Systemic Cyber (Jump-Diffusion): 1% chance of a massive "Jump"
base_cyber = np.random.lognormal(2, 0.5, n_sims)
jumps = np.random.poisson(0.01, n_sims) * 500
cyber = base_cyber + jumps
# 4. Pandemic (Poisson-Peak): High latency, high severity
pandemic = np.random.poisson(0.005, n_sims) * np.random.gamma(10, 100, n_sims)
# 5. Professional Liability (Gamma): Long-tail diversification
liability = np.random.gamma(2, 20, n_sims)
# 6. Credit & Surety (Beta): Macroeconomic cycle risk
credit = np.random.beta(0.5, 5, n_sims) * 200
```

6. Risk reinsurance

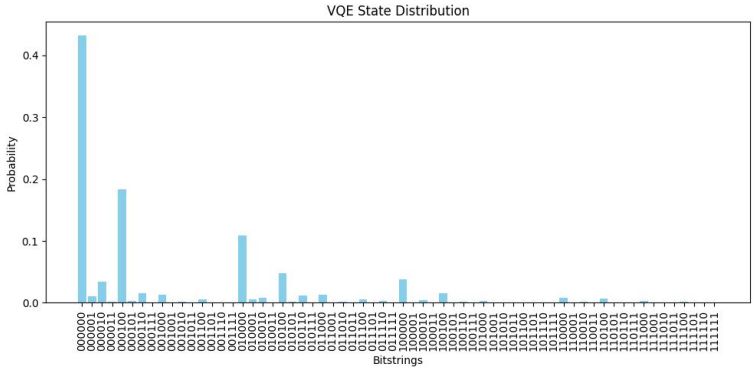
- Contracts

```
# Q0: Property XS (Wind + Flood) - Protects against natural cats
recoveries[:, 0] = np.minimum(np.maximum(windstorm + flood - 150, 0), 800)

# Q1: Cyber XS (Deep Tail) - Protects against systemic cloud failure
recoveries[:, 1] = np.minimum(np.maximum(cyber - 100, 0), 1000)

# Q2: Quota Share (QS) - 20% share of the whole 6-peril portfolio
recoveries[:, 2] = total_loss * 0.20

# Q3: Financial Lines XS (Liability + Credit) - Diversified protection
recoveries[:, 3] = np.minimum(np.maximum(liability + credit - 60, 0), 400)

# Q4: Global Stop-Loss (Low threshold) - High Capital Relief
recoveries[:, 4] = np.minimum(np.maximum(total_loss - 250, 0), 1500)

# Q5: Global Stop-Loss (High threshold) - Cheap tail protection
recoveries[:, 5] = np.minimum(np.maximum(total_loss - 500, 0), 3000)
```

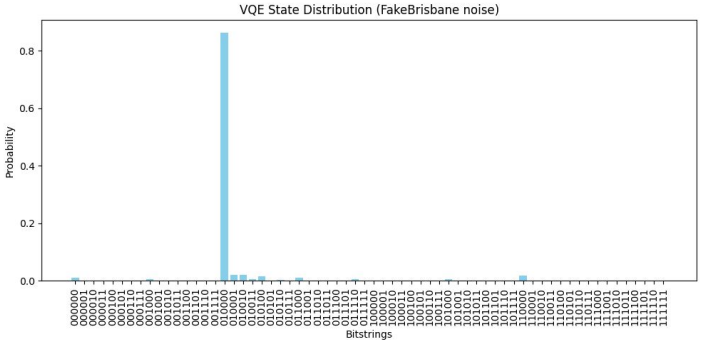
Then calculates the terms of the Hamiltonian

7. VQE results

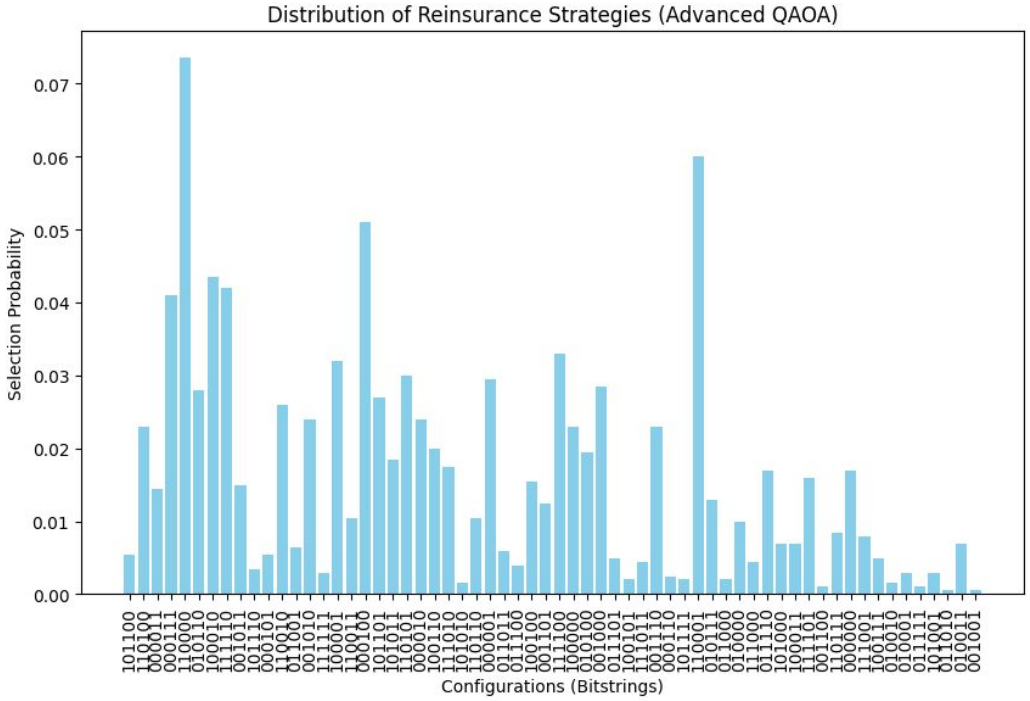
Normal VQE



meta-VQE



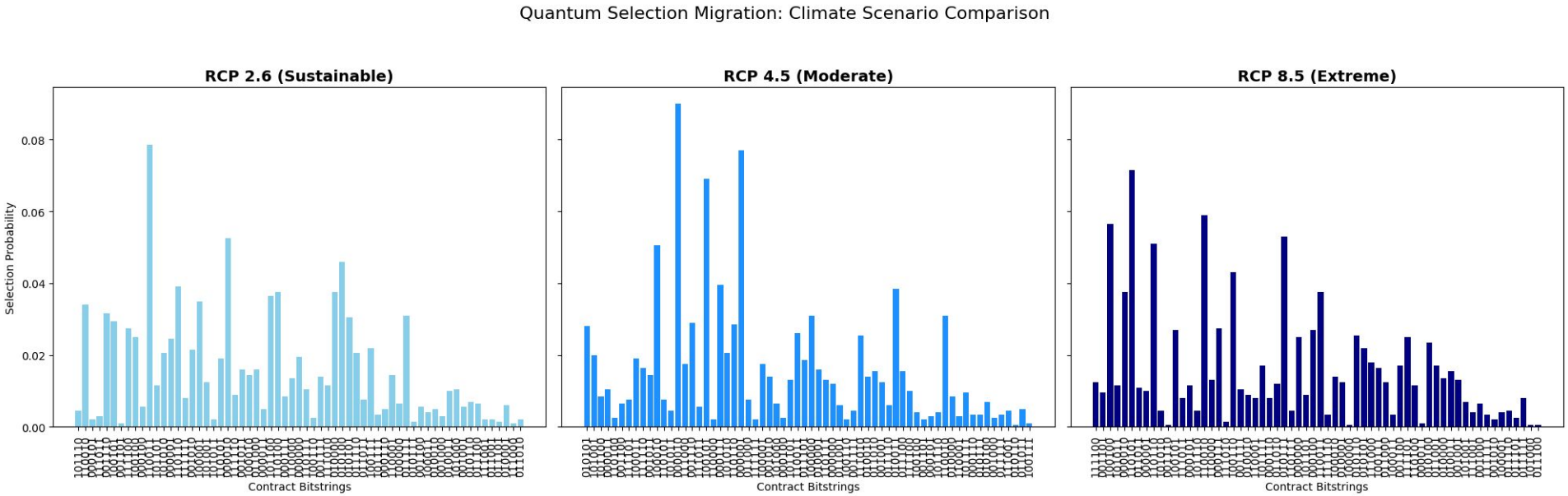
8. QAOA Results



RANG	BITSTRING	PROBA	CAPITAL LIBÉRÉ	STATUT S
#1	000011	7.3%	2000.0 M€	✅ OK
#2	100011	6.0%	2250.0 M€	✅ OK
#3	001000	5.1%	500.0 M€	✅ OK
#4	010001	4.3%	1500.0 M€	✅ OK
#5	011111	4.2%	2980.0 M€	✅ OK

9. Global warming sensitive models

Add the IPCC (GIEC) scenarios to our risk models
From left to right: Paris agreement, Intermediate, Business-as usual



10. Global warming sensitive models

Add the IPCC (GIEC) scenarios to our risk models

SCENARIO: RCP2.6

RANG	BITSTRING	PROBA	CAPITAL LIBÉRÉ	STATUT S
#1	110011	7.8%	2550.0 M€	✅ OK
#2	010000	5.2%	300.0 M€	✅ OK
#3	001010	4.6%	1300.0 M€	✅ OK
#4	011011	3.9%	2800.0 M€	✅ OK
#5	001011	3.8%	2500.0 M€	✅ OK

SCENARIO: RCP4.5

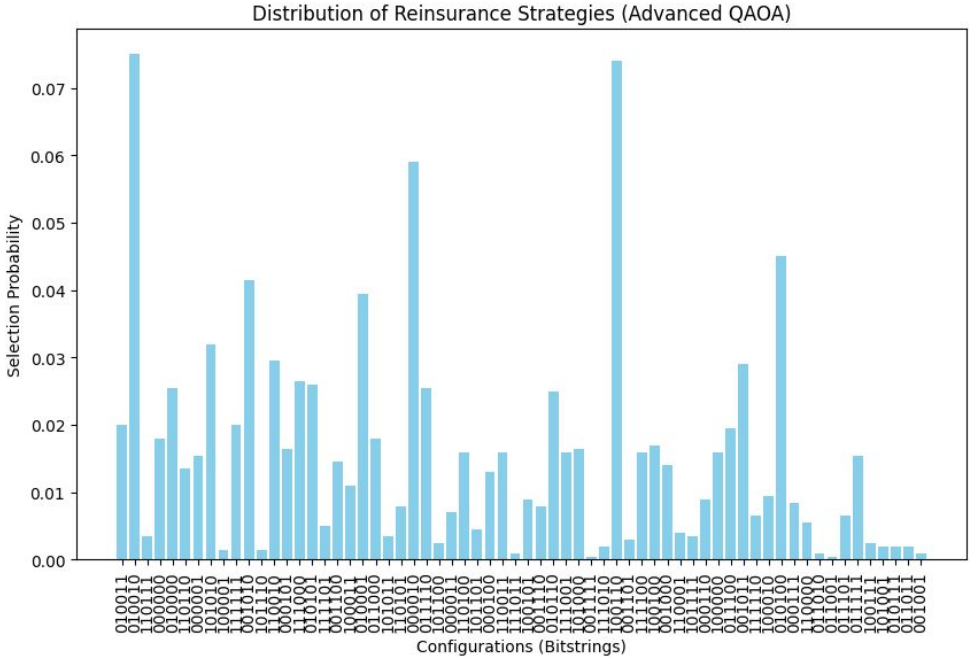
RANG	BITSTRING	PROBA	CAPITAL LIBÉRÉ	STATUT S
#1	010000	9.0%	300.0 M€	✅ OK
#2	100000	7.7%	250.0 M€	✅ OK
#3	101001	6.9%	1950.0 M€	✅ OK
#4	010001	5.1%	1500.0 M€	✅ OK
#5	111000	4.0%	1050.0 M€	✅ OK

SCENARIO: RCP8.5

RANG	BITSTRING	PROBA	CAPITAL LIBÉRÉ	STATUT S
#1	101011	7.1%	2750.0 M€	✅ OK
#2	010011	5.9%	2300.0 M€	✅ OK
#3	001001	5.7%	1700.0 M€	✅ OK
#4	111010	5.3%	1850.0 M€	✅ OK
#5	011011	5.1%	2800.0 M€	✅ OK

11. Reinsurance contracts for assets portfolios

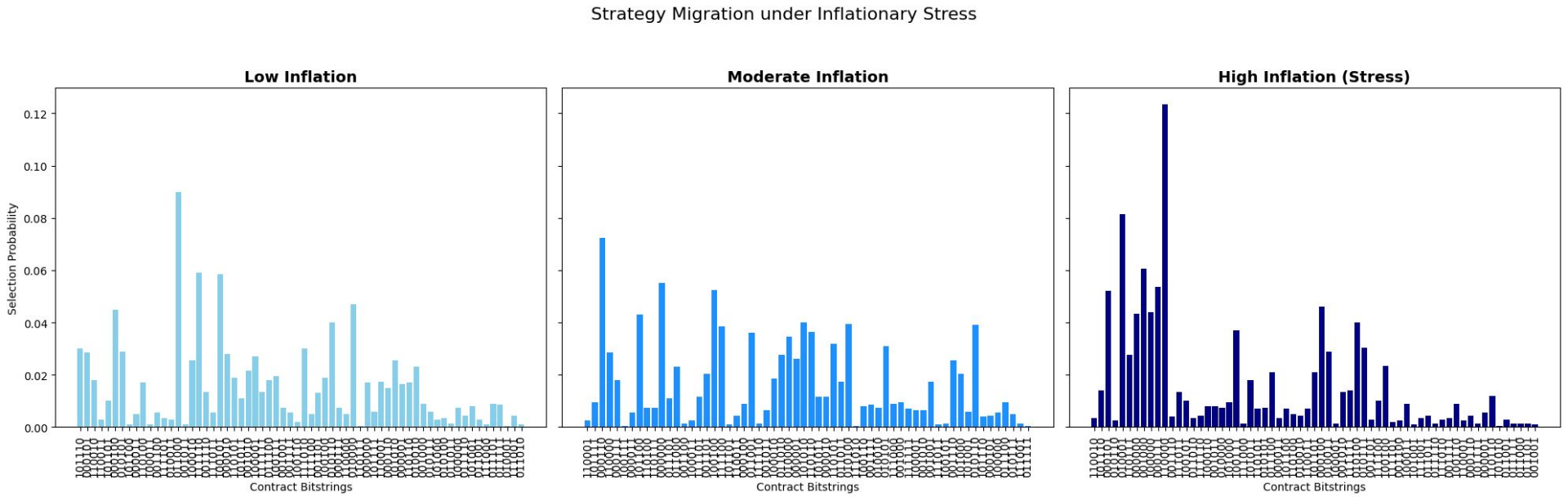
Companies taking insurances to invest into the market
Data set similar to risk dataset, but taking into account volatility, system financial contagion, black swan events, cross asset diversification…



RANG	BITSTRING	PROBA	CAPITAL LIBÉRÉ	STATUT S
#1	010010	7.5%	1600.0 M€	✅ OK
#2	011001	7.4%	3000.0 M€	✅ OK
#3	010000	5.9%	400.0 M€	✅ OK
#4	001010	4.5%	1800.0 M€	✅ OK
#5	010100	4.2%	700.0 M€	✅ OK

12. Inflation sensitive models

Add inflation scenarios to our models:
From left to right: low inflation (CPI 1-2%), moderate inflation (3-5%), high inflation (>10%)



13. Inflation sensitive models

INFLATION SCENARIO: Low

RANK	BITSTRING	PROBA	CAPITAL RELIEF	STATUS S
#1	000110	9.0%	1500.0 M€	✅ OK
#2	011111	5.9%	4500.0 M€	✅ OK
#3	101001	5.9%	3100.0 M€	✅ OK
#4	000010	4.7%	1200.0 M€	✅ OK
#5	001000	4.5%	600.0 M€	✅ OK

INFLATION SCENARIO: Moderate

RANK	BITSTRING	PROBA	CAPITAL RELIEF	STATUS S
#1	011000	7.2%	1000.0 M€	✅ OK
#2	000000	5.5%	0.0 M€	❌ FAIL
#3	001001	5.2%	2600.0 M€	✅ OK
#4	001101	4.3%	2900.0 M€	✅ OK
#5	010101	4.0%	2700.0 M€	✅ OK

INFLATION SCENARIO: High

RANK	BITSTRING	PROBA	CAPITAL RELIEF	STATUS S
#1	000011	12.3%	3200.0 M€	✅ OK
#2	100010	8.2%	1700.0 M€	✅ OK
#3	000100	6.0%	300.0 M€	✅ OK
#4	111000	5.3%	1500.0 M€	✅ OK
#5	101000	5.2%	1100.0 M€	✅ OK

Conclusion

Reinsurance contract optimisation = hard optimisation problem: QUBO approach
Cannot be simulated **classically** for big portfolios

The QAOA showed good convergence on portfolios
Difference with VQE: VQE has a tendency to give too much importance to one state (explain the curves) while QAOA explores a broader universe

Quantum computing will be useful for climatic stress tests or portfolio optimisation so there is a direct link with our work with IPCC and inflation.